

IT Research Methodology

Research Report and Evaluation

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❖ Research Report

- Research culminates in a report in different forms.
- A **research report** is a completed study that reports an investigation or exploration of a problem, identifies questions to be addressed, and includes data collected, analyzed, and interpreted by the researcher.
- Rules and procedures in place at colleges and universities govern the criteria used by faculty advisers and committees.
- Journal reviewers and journal editors employ criteria presented once a year (typically) in their journals and on the journal's website.

Audience for Research

Audiences	Standards
Faculty (adviser/ committees)	• Standards used in the past in program area • Standards traditionally used by each individual faculty member • Standards used at the college or university
Journal reviewers	• Use of published standards printed typically once for each volume and available in Author Guidelines from the journal's website • Separate standards may be published for quantitative and qualitative research • Editor must reconcile differing opinions among reviewers
Policymakers	• Ease of understanding results • Immediate application of results • Clarity and brevity of ideas
Practicing educators in the field	• Relevance of problem or issue being studied • Ease of identifying results • Practical suggestions from a research study
Conference paper reviewer	• Whether the researcher has submitted the proper materials • Whether the proposal fits the conference theme or the priorities for the conference • Whether the project summary is clearly written and organized and addresses the guidelines for a summary
The researcher	• Standards of the outlet for the study (e.g., faculty, journal, practitioners, or conferences) • Standards related to the five elements and the seven phases in the process of research • Standards related to quantitative and qualitative criteria

❖ Types of Research Reports

- Students prepare dissertations and theses as a requirement for their doctoral and master's programs
- After completing the dissertation or thesis, researchers often condense it into a journal article for submission to an educational journal
- **Dissertations and thesis** are the doctoral and master's research reports prepared for faculty and graduate committees.
- The length of a dissertation or thesis can vary, depending on the tradition in a school or department
- The doctoral dissertation might range from the shorter dissertation of 100 pages to more than 400 pages. Master's theses may run from 50 to 100 pages

❖ Types of Research Reports

- In general, qualitative studies are much longer than quantitative studies. Qualitative researchers often report many quotes and present multiple perspectives on themes.
- **A dissertation or thesis proposal** is a plan for a research report, initiated and developed
- before the research actually begins.
- One essential difference between proposals and reports is that you will write a proposal before conducting the research.
- **In a proposal**, the researcher writes **about what will take place**; in the final study, the investigator writes about what has taken place. This means that a proposal will likely be written using a future verb tense (i.e., will) to describe the action.

❖ Types of Research Reports

- **In a final study**, the past verb tense (i.e., was) describes what has already occurred. However, when the proposal contains a pilot or preliminary study, you need to describe it as an action that has already occurred.
- A well-defined proposal does the following:
  - Facilitates the process of obtaining permissions to study a site or educational setting.
  - Provides information to gatekeepers and those in authority so that they can determine the likely impact of a study at their site.
  - Provides criteria to assess the quality of a project. Those evaluating and reviewing a study use these criteria. Knowing the proper elements of a good proposal permits evaluators to examine projects for these elements and to determine, once you complete a project, whether it fulfills its goals.

Quantitative and Qualitative Dissertation and Thesis Proposals

Formats of Quantitative and Qualitative Proposals

Quantitative Format	Qualitative Format
Title Page	Title Page
Abstract	Abstract
Introduction	Introduction
Statement of the Problem	Statement of the Problem
Purpose and Research Questions or Hypotheses	The Purpose and Research Question
Theoretical Perspective	Delimitations and Limitations
Definition of Terms	Procedure
Delimitations and Limitations of the Study	Qualitative Methodology and Design
Review of the Literature	Research Site and Purposeful Sampling
Methods	Data Analysis Procedures
Study Design	Researcher's Role and Potential Ethical Issues
Procedures, Instruments, Reliability, Validity	Methods of Validation
Data Analysis	Preliminary Findings
Preliminary Results	Anticipated Outcomes of the Study and Tentative
Potential Ethical Issues	Literature Review (Optional)
Time Line, Budget, and Preliminary Chapter Outline	Time Line, Budget, and Preliminary Chapter Outline
References	References
Appendices	Appendices

❖ Journal Articles and Conference Papers

- A norm of research is that individuals publicly share their research reports so that information is accessible to the general education community.
- **A journal article** is a polished, shorter research report that you send to an editor of a journal. The **editor** arranges for **two or three reviewers** to provide comments about the study.
- The editor then makes a decision based on the reviewers' comments that typically falls into one of three categories: **accept, revise and resubmit, or reject**.
- Usually, when the reviewers accept the manuscript provisionally, they do so based on their anticipation of successful revisions by the author.
- A journal article is much shorter than a thesis or dissertation because of page limitations imposed by publishers and editors of journals.

❖ Journal Articles and Conference Papers

- **A conference paper** is a research report presented to an audience at a state, regional, national, or international conference typically sponsored by a professional association.
- Developing and presenting a conference paper from a research study help to publicize the study, provide a research entry for a résumé, help to advance the work of the author within the educational research community, and build knowledge among researchers exploring the topic.
- Researchers prepare conference papers for audiences who attend the conference as well as for individuals who review and accept the papers for presentation.

❖ Journal Articles and Conference Papers

➢ A summary about your research that includes the following:

- ✓Objectives or purposes
- ✓Perspective(s) or theoretical framework
- ✓Methods, techniques, or modes of inquiry
- ✓Data sources, evidence, objects, or materials
- ✓Results and/or conclusions/point of view
- ✓Scholarly or scientific importance of the study

Design an Appropriate Quantitative Structure

The Front Matter, Body, and Back Matter of a Quantitative Study

Front Matter
Title page
Abstract of the study (optional)

Body of the Paper
Introduction
- Statement of the problem
- Purpose statement
- Research questions or hypotheses
- Theoretical or conceptual explanation

Review of the Literature
- Review of the previous research
- Summary of major themes
- How present study will extend literature

Methods
- Sample and site
- Access and permissions
- Instruments and their reliability and validity
- Interventions (if used)
- Procedures of data collection
- Analysis of the data

Results
- Descriptive analysis of all data
- Inferential analysis to address questions/hypotheses
- Tables and figures to display the data

Discussion
- Summary of major results
- Relationship of results to existing studies
- Limitations of the study
- Implications for future research
- Overall significance of the study

Back Matter
References
Appendices (e.g., instruments, key questions)

Design an Appropriate Qualitative Structure

The Front Matter, Body, and Back Matter of a Scientific Qualitative Structure

Front Matter
Title page
Preface and acknowledgments (optional)
Table of contents (optional)
List of tables (optional)
List of figures (optional)
Abstract of the study (optional)

Body of the Paper
Introduction
- Statement of the problem
- Purpose statement
- Research questions

Procedures
- Rationale for qualitative approach
- Sample and site
- Access and permissions
- Data-gathering strategies
- Data analysis approach

Findings
- Description of site or individuals
- Analysis of themes

Discussion
- Major findings
- Comparison of findings with existing studies
- Limitations
- Implications for future research
- Overall significance of the study

Back Matter
References
Appendices (e.g., figures, interview, or observational protocols)

❖ A Sensitive, Ethical, And Scholarly Way

➢ In addition to understanding the structure of a study, researchers engage in good writing practices when they compose a research report:

- ✓Are sensitive to individuals and use language that reduces bias
- ✓Use appropriate research terms
- ✓Write and report the findings ethically
- ✓Employ a point of view consistent with quantitative and qualitative approaches
- ✓Balance research and content
- ✓Interconnect parts of a study
- ✓Advance a concise title

Quantitative Criteria for Evaluating Quantitative Research

Research Problem:
- Is it stated?
- Is it clear?
- Is it complete and accurate?
- Does it offer theoretical and practical value?

The Literature Review:
- Is it clear, relevant, recent, and based on studies reporting numbers?
- Are the citations and references accurate?

Hypotheses and Variables:
- What are the hypotheses, and what types are they?
- Does the study indicate the independent, dependent, intervening, moderating, and control variables?
- Were operational definitions given for the variables?
- Were extraneous effects controlled in the study so that bias did not have an effect on the participants, their experiences, or the generalization of the results?
- Were certain variables manipulated?

Design:
- Was the study design identified?
- Were the scores from the measures valid and reliable?
- Were the statistics the right choice?

Findings and Discussion:
- Did the findings fit the problem?
- How significant and important were the findings?
- Did the discussion section report conclusions, and were they consistent with the study's results?
- Did the discussion section offer reasonable interpretations of why results did and did not match expectations?
- Did the discussion section suggest reasonable implications about what readers should do with the results?

Quantitative Criteria for Evaluating Quantitative Research

Title for the Study
_____ Does it reflect the major independent and dependent variables?
_____ Does it express either a comparison among groups or a relationship among variables?
_____ Does it convey the participants and site for the study?

Problem Statement
_____ Does it indicate an educational issue to study?
_____ Has the author provided evidence that this issue is important?
_____ Is there some indication that the author located this issue through a search of past literature or from personal experiences?
_____ Does the research problem fit a quantitative approach?
_____ Are the assumptions of the study consistent with an approach?

Review of the Literature
_____ Are the studies about the independent and dependent variables clearly reviewed?
_____ Does the review end with how the author will extend or expand the current body of literature?
_____ Does the study follow APA style?

Purpose, Hypotheses, and Research Questions
_____ Does the author specify a purpose statement?
_____ Is the purpose statement clear, and does it indicate the variables, their relationship, and the people and site to be studied?
_____ Are either hypotheses or research questions written?
_____ Do these hypotheses or questions indicate the major variables and the participants in a study?
_____ Do the purpose statement and hypotheses or research questions contain the major components that will help a reader understand the study?
_____ Has the author identified a theory or explanation for the hypotheses or questions?

Data Collection
_____ Does the author mention the steps taken to obtain access to people and sites?
_____ Is a rigorous probability sampling strategy used?
_____ Has the author identified good, valid, and reliable instruments to use to measure the variables?
_____ Are the instruments administered so that bias and error are not introduced into the study?

Data Analysis and Results
_____ Are the statistics chosen for analysis consistent with the research questions, hypotheses, variables, and scales of measurement?
_____ Is the unit of analysis appropriate to address the research problem?
_____ Are the data adequately represented in tables and figures?
_____ Do the results answer the research questions and address the research problem?
_____ Are the results substantiated by the evidence?
_____ Are generalizations from the results limited to the population of participants in the study?

Writing
_____ Is the structure of the overall study consistent with the topics addressed in a quantitative study?
_____ Are educational and social science terms carefully defined?
_____ Are variables labeled in a consistent way throughout the study?
_____ Is the study written using extensive references?
_____ Is the study written using an impersonal point of view?
_____ Is the study written appropriately for intended audience(s)?

Q&A